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An Interpretable Statistical Surrogate of Activated Sludge Systems that Preserves Mass Conservation and Component Non-Negativity --Manuscript Draft--

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Abstract:	Activated sludge models are essential for wastewater-treatment design and operation, but their detailed simulations can be too computationally demanding for repeated analysis. Machine-learning surrogates are faster, yet accurate predictions may still violate mass conservation or produce negative component concentrations. This study introduces Invariant-Constrained Second-Order Regression (ICSOR), an interpretable surrogate that combines a transparent, invariant-aware regression head with a constrained deployment procedure. The regression head predicts the effluent component state from influent conditions and operating inputs. At deployment, each prediction is checked, projected onto the mass-conservation constraints, and, when necessary, adjusted by a linear program to enforce both conservation and non-negativity. ICSOR was evaluated on 10,000 steady-state samples from an ASM2d-TSN continuous stirred tank reactor and compared with XGBoost, LightGBM, CatBoost, AdaBoost, Random Forest, support vector regression, k -nearest neighbors, partial least squares, and a multilayer perceptron. The deployed ICSOR predictions had zero conservation and non-negativity violations: the affine projection resolved 21.8% of test cases, and the linear program resolved the remaining 78.2%. For context, none of the raw regression-head predictions satisfied conservation, although 21.7% were non-negative, showing that the hard guarantee came from constrained deployment rather than training alone. This physical reliability involved a moderate accuracy tradeoff: aggregate test RMSE was 5.98, compared with 4.38 for the multilayer perceptron and 5.30 for LightGBM. Across the dataset-size study, however, ICSOR achieved the lowest normalized area under the RMSE learning curve and the smallest train-validation RMSE gap among the comparatively accurate models. ICSOR therefore offers a useful balance of interpretability, data efficiency, and guaranteed physical admissibility for steady-state activated sludge surrogate modeling.
Additional Information:	
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